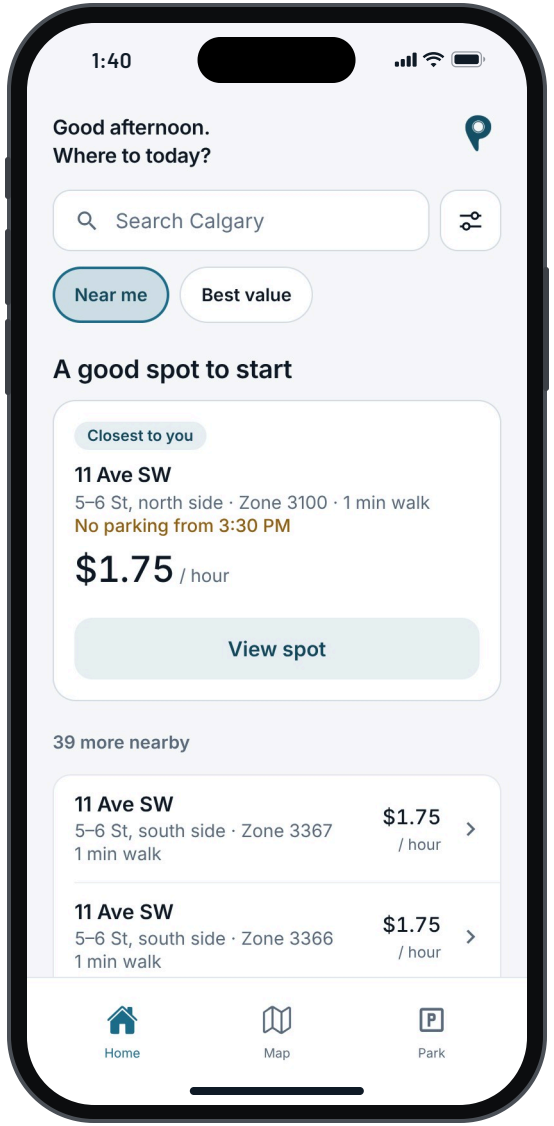
CASE STUDY

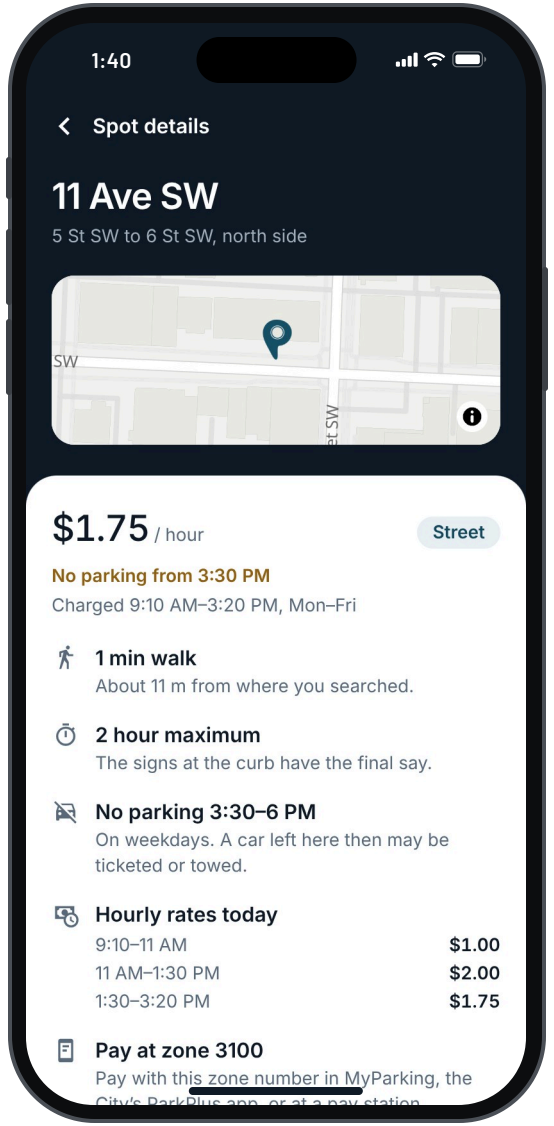
ParkAid

A mobile app for finding street parking in downtown Calgary, built on the City of Calgary's open data for parking zones, rates and restrictions.

PROJECT DETAILS

ROLE	Team member in a five-person SAIT capstone. Afterward, I continued the project on my own, with AI coding tools.
TIMELINE	Capstone in Year 2, Semester 2. Follow-up work from September 2025 to November 2026.
PLATFORMS	iOS, Android and web from one React Native (Expo) codebase.
BUILT WITH	React Native, Expo, MapLibre GL JS, react-native-maps, Node.js, Express, PostgreSQL with PostGIS, Jest.





Ehrl Balquin · linkedin.com/in/ehrlbalquin · github.com/ginesbal/parkaid

NEXT 02 · SUMMARY →

The project in short

The problem

The capstone version showed prices that did not match the City's rates, used raw City addresses, and listed the same block several times.

My role

One of five in the capstone. Afterward I continued alone. Claude Code, an AI coding tool, wrote most of the code from March 2026, from my instructions.

What changed

The app reads the City's rates by time of day and its rush-hour rules, shows each block once, and names it by block, side of the street and zone.

Results

Rates and rules match the City's data for the tested blocks. 45 overlapping pins became 15. 170 automated tests pass. No usability testing yet.

CONTENTS: ONE STALL PER PAGE

03 BEFORE AND AFTER
The Home screen, capstone and current

04 FOUR QUESTIONS
What a driver needs to know at the curb

05 THE MAP
45 parking blocks within 250 m

06 PARKING A CAR
Length, cost and paying

07 HOW IT IS BUILT
System, my front-end work, the hard parts

08 HOW I WORKED
Role, AI tools and testing

09 LINKS
Source code, demo, Figma and next steps

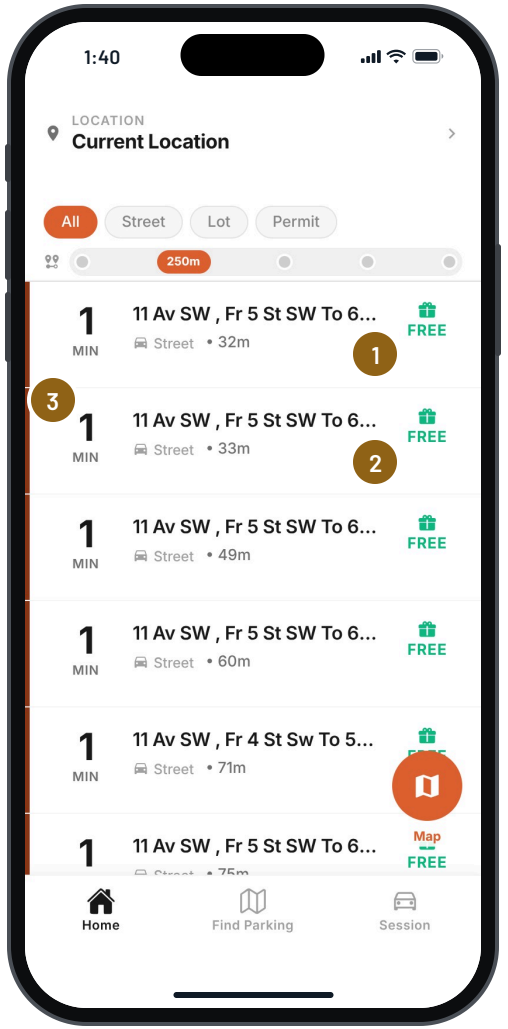
The Home screen, capstone and current

The same blocks on the same Tuesday at 1:40 PM. Left: the capstone code from October 2025. Right: the current version.

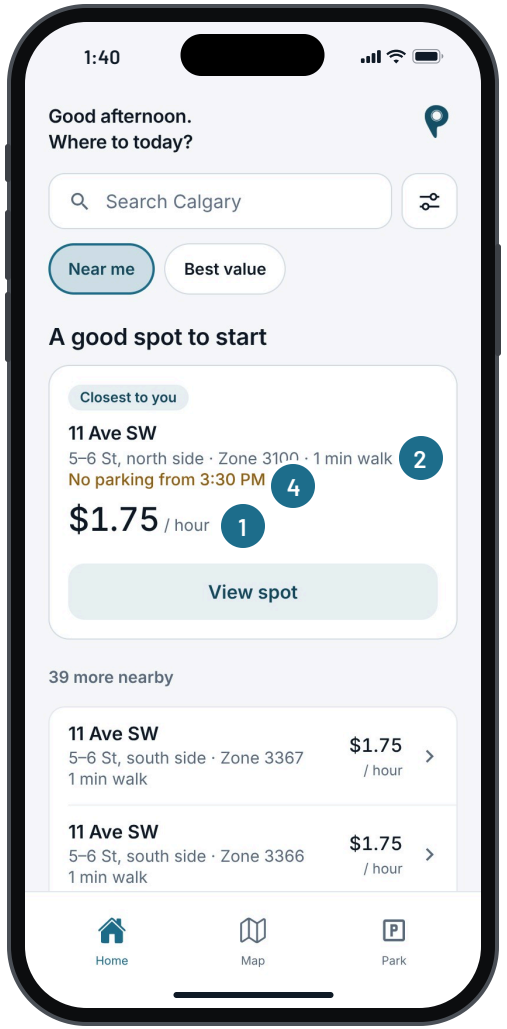
		CAPSTONE	CURRENT
1	**Price**	FREE, from a lookup on the zone code	$1.75 an hour, the City's rate now
2	**Location**	"11 Av SW , Fr 5 St SW To 6 St SW"	"5–6 St, north side · Zone 3100"
3	**Rows**	One row per City segment, so blocks repeat	One row per block: 277 of 1,178 rows were repeats
4	**Rules**	Not shown	"No parking from 3:30 PM"

Numbers match the markers on the screens. Row 3 has no marker on the current screen: the repeats are gone.

CAPSTONE



CURRENT



What a driver needs to know at the curb

The Spot details screen answers four questions for one block: 11 Ave SW, between 5 St and 6 St, north side.

1 Can I park here now?

Yes, until 3:30 PM. From 3:30 to 6 PM on weekdays, nobody may park on this block.

2 What does it cost right now?

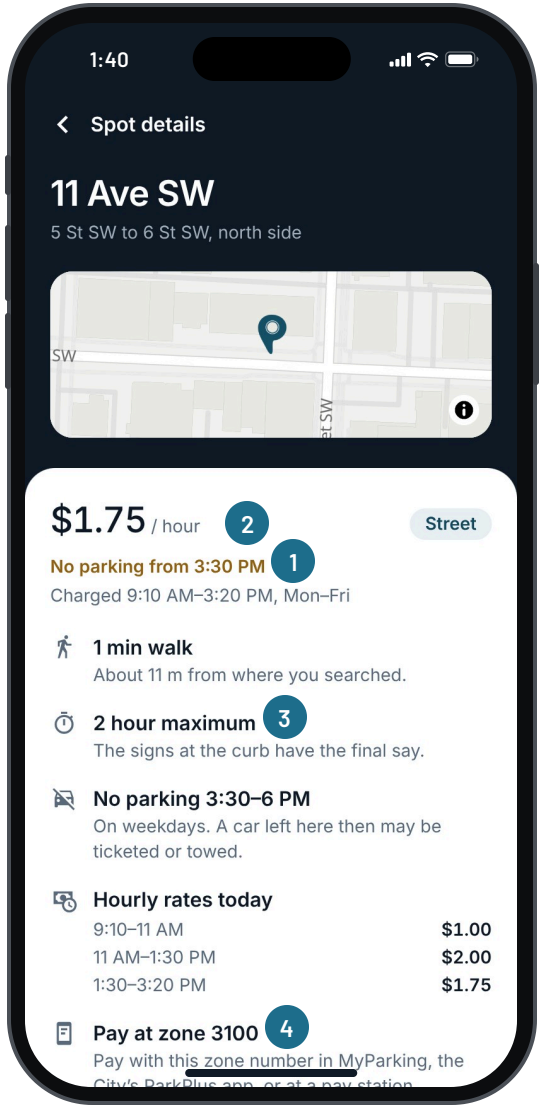
$1.75 an hour. The rate changes at 11 AM and 1:30 PM, and the day's rates are listed further down.

3 How long can I stay?

Two hours at most. The app adds that the signs at the curb have the final say.

4 Which zone do I pay?

Zone 3100. One tap copies it for MyParking, the City's ParkPlus app.



45 parking blocks within 250 m

Downtown, the pins pile up. Pins that would touch are grouped, so every pin on the map can be read.

1. **Group pin.** How many spots it holds and the lowest rate. A tap zooms in until they separate.
2. **Price pin.** One spot and its rate right now. The selected pin turns dark.
3. **Sheet.** The suggested spot. “Show all 44 spots” opens the list.

Sorting

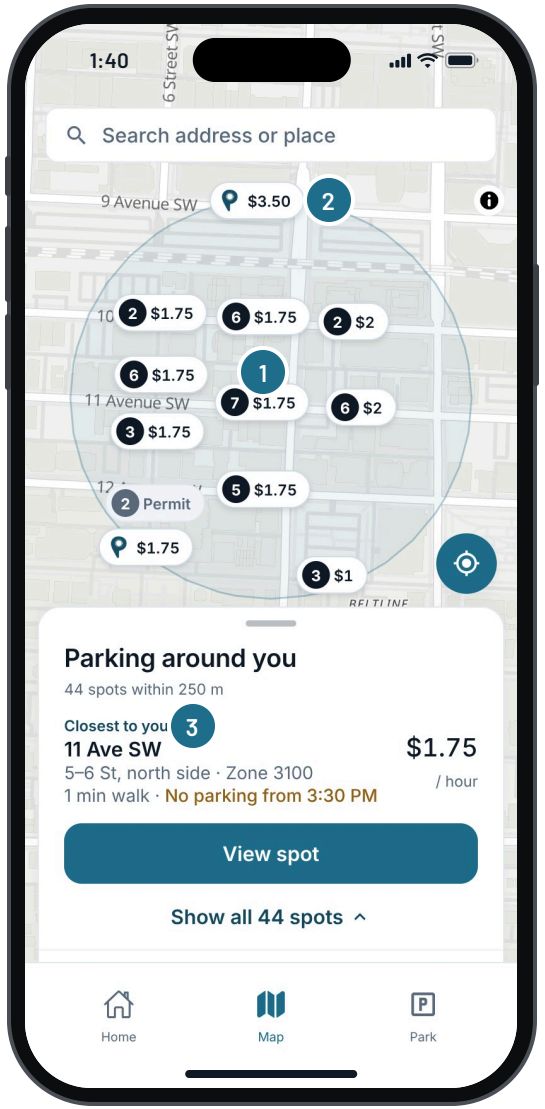
Near me: walking time, then price. **Best value:** rate now, then distance.

Filters

Street, lot, parkade or permit, within 150 m to 1 km. The list and the map share them.

The grouping rule. Pins closer than a group pill (70 × 34 px at the current zoom) share one pin, led by the best-ranked spot. If every spot in a group is within 12 m, a tap selects it instead of zooming. The same code runs on Google Maps and on MapLibre.

Measured on the opening web view: 45 pins with 44 overlapping became 15 pins with 4 touching.



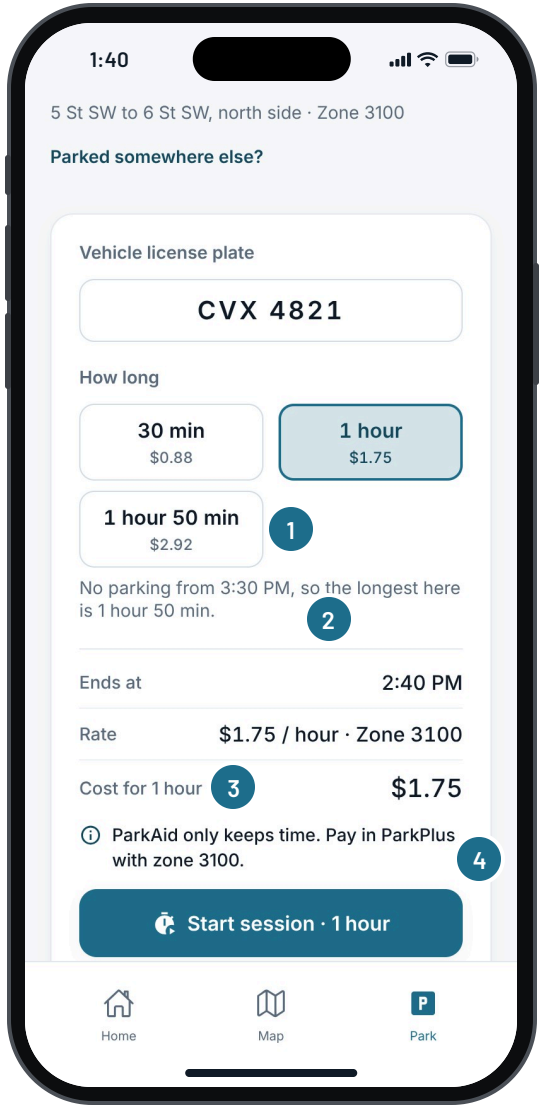
Length, cost and paying, before the session starts

The Park screen for the same block at 1:40 PM. No parking starts at 3:30 PM.

1. **Lengths stop at the rule.** The longest choice is 1 hour 50 min, so the car is gone by 3:30 PM.
2. **The reason is on screen.** “No parking from 3:30 PM, so the longest here is 1 hour 50 min.”
3. **Cost before starting.** Ends at 2:40 PM and costs $1.75, at the City’s rate.
4. **Paying.** ParkAid only keeps time. The driver pays in ParkPlus with zone 3100, and is reminded to end that session too.

Later the same day. At 2:40 PM the longest stay on this block is 50 min. At 3:40 PM the list says “No parking until 6:00 PM”, and Spot details offers no Start. Both are in the rush-hour recording.

The capstone version had a demo here: the driver picked a rate of $1.50, $2.50 or $3.50 an hour.



One codebase for iOS, Android and web

CITY OF CALGARY OPEN DATA
Zones, rates, restrictions

→

IMPORT SCRIPT
Four datasets into one table

→

POSTGRESQL + POSTGIS
Supabase; finds nearby spots

→

EXPRESS API
Render; merges block segments

→

REACT NATIVE APP
Google Maps on iOS and Android, MapLibre on the web

My front-end work

- **Screens:** Welcome, Home, Map, Spot details, Park and Session, with React Navigation.
- **Components:** spot rows, map and group pins, payment actions.
- **Rules in tested functions:** rates, rush hour, ranking, grouping.
- **Platform code:** a web map file, a smaller pin for Android.
- **State:** a session context, hooks for location and spots, cached responses.
- **Accessibility:** screen reader labels, Reduce Motion, keyboard order on the web.

The hard parts

Rates change during the day	The City's rate bands become time ranges; costs count only charged hours.
Rush-hour rules	Checked against the time now and the end of a planned stay.
Repeated segments	The API merges rows for the same stretch of a block.
Crowded map	Pins grouped by distance on screen, on both map libraries.
Android markers	Drawn into 100 × 100 px images, which cut off labels; Android gets a smaller badge.

Expo SDK 54 · React Native 0.81 · React Navigation · react-native-maps · MapLibre GL JS · Node.js · Express · PostgreSQL + PostGIS on Supabase · Render · Jest

My role, AI tools and testing

Role

In the SAIT capstone, a team of five built the base app: list, map, session screen, API and data import. This repo starts from that code (September 29, 2025), so the team's commits are not in it. After the capstone, I continued on my own.

AI tools

Up to October 28, 2026, Claude Code wrote 101 of the 128 commits, from my instructions. 27 are under my accounts; a few came from an OpenAI Codex branch. I decided what to build and fix, and in what order.

Each review round

1. **Capture:** the web build with City data at a fixed day and time
2. **Review:** usability, wording, contrast, keyboard, screen sizes
3. **Decide:** I chose the order: rates and rules first
4. **Fix:** one change per commit
5. **Verify:** tests, three screen widths, Android build

What was tested

Unit tests	152 app and 18 backend tests pass
Real data	1,672 City records near 11 Ave SW and 5 St SW
Map density	45 pins (44 overlapping) to 15 (4 touching)
Contrast	Gray on selected rows 4.23:1 to 4.74:1; AA is 4.5:1
Keyboard (web)	Pins skipped, the list has them; search first
Small screens	At 360 × 640, Add time and End session now on screen
Builds	Web at 360, 390, 1280 px; Android bundle

Not tested yet: usability with drivers, a physical phone, the database test suites, query speed.

See it, read the code, open the design

SOURCE CODE
github.com/ginesbal/parkaid →

DEMO RECORDINGS
Seven clips of the real build · github.com/ginesbal/parkaid/tree/main/docs/walkthrough →

FIGMA FILE
Variables, icons, 11 components, 5 screens · figma.com/design/dLkYCXg0Xzfm4NAGWAwcJF →

Recordings: 1 Finding parking, 2 The map, 3 Spot details, 4 Parking, 5 Rush hour, 6 Details while parked, 7 On a laptop. The Figma file is rebuilt from the code; its session screens and prototype links are not finished.

Next steps

- Usability tests with five drivers downtown.
- Testing on a physical iPhone and Android phone.
- Loading the new data import into the live database.
- Finishing the Figma file.

About the screens

The current screens are the app's web build in an iPhone-sized window (390 × 844 pt) with iPhone safe areas emulated. The capstone screen is its October 2025 web build, with a plain status bar added above it. All show the City's records on a Tuesday at 1:40 PM. The phone frames and status bars are drawn on.

This case study was drafted with Claude Code. Its numbers come from the repository and from test runs.